

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{ccc} \psi^\dagger \psi & \Rightarrow & \beta \int_{\alpha_-}^{\alpha_+} \frac{dp}{2\pi} = \frac{\beta}{2\pi} (\alpha_+ - \alpha_-) \\ \frac{1}{\beta^2} \frac{\partial \psi^\dagger}{\partial \lambda} \frac{\partial \psi}{\partial \lambda} & \Rightarrow & \frac{\beta}{2\pi} \int dp p^2 = \frac{\beta}{6\pi} (\alpha_+^3 - \alpha_-^3) \\ \dots & \dots & \dots \end{array}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} x &= \varepsilon^{-1} (R_n R_{n+1}) \log^{2N-3} \left(\left(\frac{R_n R_{n+1}}{\varepsilon} \right)^{\frac{R_n R_{n+1}}{2}} \frac{N-1}{\eta} \right), \\ y &= \log^4 \left(\left(\frac{R_n R_{n+1}}{\varepsilon} \right)^{\frac{1}{2}} \log^{N-1} \left(\left(\frac{R_n R_{n+1}}{\varepsilon} \right)^{\frac{R_n R_{n+1}}{2}} \frac{N-1}{\eta} \right) \right) \log \prod_{j \neq n} I_j, \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} C_n(x) &= \min \left\{ \begin{array}{l} L_n(x) + \int_0^\infty C_{n-1}(x-\xi) f_n(\xi) d\xi, \\ \min_{x < y \leq x+B} \{ K + v(y-x) + L_n(y) + \int_0^\infty C_{n-1}(y-\xi) f_n(\xi) d\xi \} \end{array} \right\}, \\ G_n(x) &= vx + L_n(x) + \int_0^\infty C_{n-1}(x-\xi) f_n(\xi) d\xi, \\ C_n(x) &= -vx + \min \{ G_n(x), K + \min_{x \leq y \leq x+B} G_n(y) \}. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\left\{ \begin{array}{ll} j^\mu(x) = \lim_{y \rightarrow x} \frac{1}{2} (\bar{\psi}(x) \gamma^\mu \psi(y) + \bar{\psi}(y) \gamma^\mu \psi(x)) = -\frac{\beta}{2\pi} \epsilon^{\mu\nu} \partial_\nu \phi(x) \\ \text{beginalign*3mm} \quad \frac{4\pi}{\beta^2} = 1 + g/\pi \\ \text{beginalign*3mm} \quad \frac{m_0^2}{\beta^2} \cos \beta \phi = ZM \bar{\psi} \psi \end{array} \right. ,$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{ccc} \{U_\alpha^1\} & \xleftarrow{\rho_{V,\{U_\alpha^1\}}} & \{U_\alpha^1|_V\} \\ \rho_{12} \uparrow & & \uparrow \rho_{V,12} \\ \{U_\alpha^2\} & \xleftarrow{\rho_{V,\{U_\alpha^2\}}} & \{U_\alpha^2|_V\} \end{array}$$